

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459913.7196 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1594 +/- 0.0063
Transit depth (Rp/Rs)^2	2.54 +/- 0.2 [%]
Orbital Inclination (inc)	89.85 +/- 0.89
Airmass coefficient 1 (a1)	1.1784 +/- 0.0056
Airmass coefficient 2 (a2)	-0.144 +/- 0.022
Scatter in the residuals of the lightcurve fit is	0.48 %
Best Comparison Star	None
Optimal Aperture	4.57
Optimal Annulus	11.09
Transit Duration (day)	0.13167 +/- 0.00082

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1594 ± 0.0063 vs 0.1489 (deviation 1.54σ)
Duration fit vs published	0.13167 d vs 0.1304 d (deviation 1.42σ)
Mid-time O–C	2.5 min
Depth SNR	23.4
Residual scatter	0.48 %

Light curve

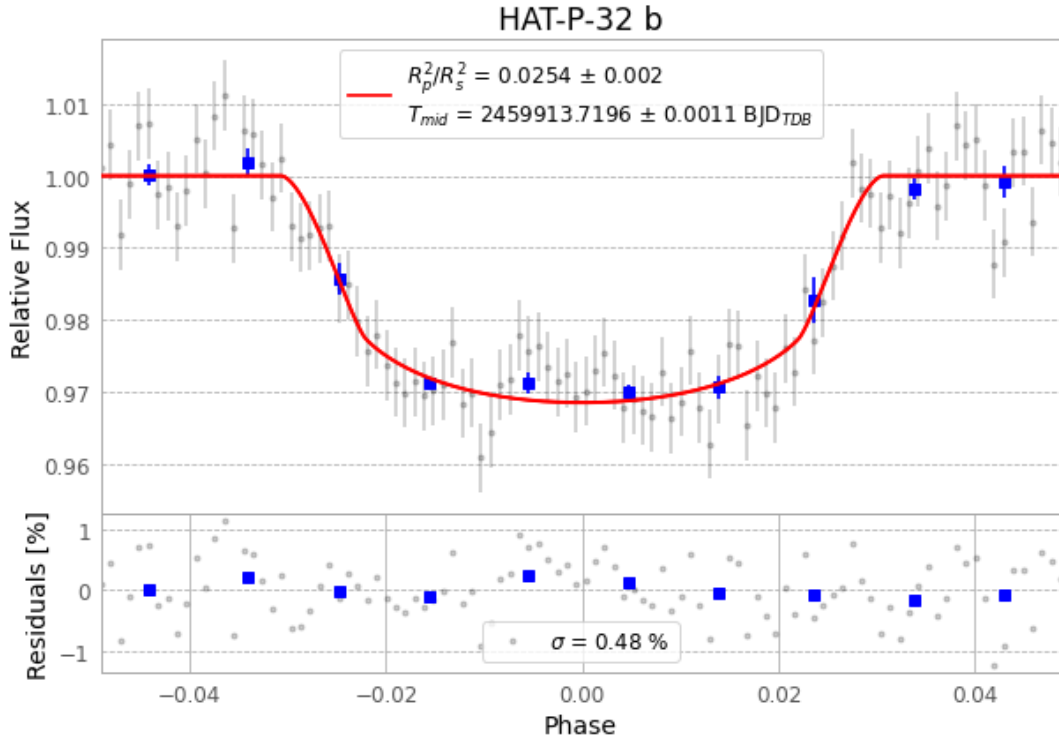


Figure 1 — FinalLightCurve_HAT-P-32 b_2022-11-29.png (SHA-256 d3fc69b55a81a387...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [524, 129], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c3172634a904c4ef...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit f0dcec7

Data provenance

102 FITS frames (67 MB), HATP-32221130023615.FITS → HATP-32221130073915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-221130.log (ba473d5d92ff...), FinalLightCurve_HAT-P-32 b_2022-11-29.png (d3fc69b55a81...), FinalLightCurve_HAT-P-32 b_2022-11-29.csv (b0c4452c47e0...)

Compute resources

Wall time 105.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 04:41Z.